

From One Algorithm to One Evidence Standard

A Federal Benchmark-and-Disclosure Proposal

B.02 — Advocacy Paper

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Abstract

The original version of this Article proposed that Congress mandate APPORTIONREGIONS as the final-map algorithm by extending the Huntington–Hill analogy inside States. Subsequent algorithm, legal, community, and evidence review showed that this framing concealed rather than eliminated value judgments. Structure, weights, resolution, seed, data corrections, VRA obligations, and community criteria all require governance.

This revised Article proposes a narrower federal rule: every State must publish a precommitted, partisan-input-excluded geographic benchmark and a complete baseline-to-final decision record. The legally authorized State body or court retains responsibility for the final plan. VRA-required changes are mandatory; State-law, community, and error corrections are permitted only through a publicly defined authority, alternatives, effects, and reasons.

The canonical benchmark uses census blocks, standard recursive bisection, shared-boundary weights, and one manifest-derived seed. APPORTIONREGIONS remains a

research comparator, not a constitutionally compelled method. The central federal claim is reproducibility and disclosure, not a uniquely fair output.

Evidence and policy status. This paper is a short internal policy explainer, not a complete law-review constitutional analysis. It is not the canonical national-standard specification, enacted legislation, external constitutional review, or a completed 50-state seed-stability certificate. The controlling candidate decision record is docs/specs/2026-07-09-national-redistricting-sta

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THE REVISED LEGISLATIVE THESIS

Rucho v. Common Cause identifies a federal constitutional justiciability limit, not a ban on positive legislation [ruc, 2019]. Congress can require an auditable congressional-election process without selecting one partisan metric or requiring one final map.

The proposed rule has four layers:

1. a constitutional and data floor;
2. a reproducible geographic benchmark;
3. a governed final-plan modification process; and
4. multi-metric evaluation that does not automatically select the map.

WHY THE BENCHMARK IS NOT THE FINAL MAP

A mandatory algorithmic final map would turn one geometric preference into a national substantive rule. It would also treat VRA, State-law, community, and topology corrections as departures from legality rather than part of legality.

The benchmark instead creates a common evidentiary anchor. A State may depart, but it must identify every moved block, authority, alternative, effect, public comment, and reason. A court can review whether the published process occurred without assuming that the benchmark itself is lawful or fair.

CANONICAL PROFILE

This profile differs from the original APPORTIONREGIONS proposal. Prime-factor structure, county weighting, and convergence search remain testable research choices. They are not consequences of constitutional text.

Table 1: Candidate national benchmark profile.

Component	Rule
Unit	Census block
Structure	Standard floor/ceiling recursive bisection
Weights	Shared geographic-boundary length
Search	One precommitted seed
Seed	SHA-256 of the complete canonical input manifest
Population	Equality as nearly exact as practicable
Political/racial/COI inputs	Excluded from benchmark generation

FINAL-PLAN AUTHORITY

The final plan must change only to the extent required by the Constitution, the Voting Rights Act, or a court order. Under *Allen v. Milligan*, the race-excluded benchmark is not the appropriate vote-dilution baseline and creates no VRA presumption, intent inference, or defense.

A final plan may also implement controlling State law, a precommitted community-of-interest criterion, or a documented data/topology correction. The record must include precommitted departure measures and a reproducible lower- departure lawful alternative. Federally required remedies are compared with a fully compliant alternative, not with proximity to the benchmark.

Where race-conscious and State-authorized partisan considerations affect overlapping districts, the record must separate each authority, input, stage, decision maker, and effect. Separate software runs alone do not establish disentanglement. Partisan evidence used for racially polarized voting analysis is not prohibited optimization input.

ADMINISTRATION

Congress owns assignment-affecting statutory choices. Census owns source data and geographic corrections. NIST owns schemas, canonicalization, and technical conformance. The Election Assistance Commission owns publication, State assistance, public comments, and challenge intake. A balanced advisory board includes State, technical, civil-rights, community, civic, and cross-partisan roles but cannot amend the benchmark.

Publication should use a certified sequential-cut ceremony when exact split evidence is available. Before the root cut, the administering body publishes a genesis commitment to the rule profile, root instance, canonical district tree, review interval, witness policy, and challenge rules. Later cuts are released by tree depth only after the prior review window. Independent observers can verify each disclosed prefix without receiving future descendants. A sustained challenge preserves a halted transcript and requires an explicitly linked new ceremony rather than silent substitution.

This publication layer does not transfer final-plan authority to software. Legally required changes and governed departures remain the responsibility of the authorized body and must be recorded outside the frozen benchmark. External transparency-log or trusted-timestamp validation is necessary before statutory deadlines can rely on ceremony timestamps.

CONSTITUTIONAL POSTURES

The primary posture is an Elections Clause disclosure and process floor, with States retaining final line-drawing authority. A federal service can publish the benchmark when a State does not. Grants support commissions and courts. A conditional-funding alternative remains severable if direct State duties are invalidated.

These alternatives are not equivalent. The direct posture maximizes uniformity; the grant and funding postures reduce commandeering risk but permit uneven participation. The tradeoff belongs in legislative deliberation.

The principal adverse authorities are the anti-commandeering rules of *Printz v. United States* and *Murphy v. NCAA*, Article III standing limits under *TransUnion LLC v. Ramirez*, and the anti-coercion limit on conditional spending under *NFIB v. Sebelius*. The model bill therefore pairs direct duties with a federal benchmark service, a concrete- injury standing rule, an interim-plan remedy, and a separately operative prospective grant condition. Whether those provisions survive review is an open legal question, not a software conclusion.

EVIDENCE STATUS

BISECT now implements bounded exact split certificates, recursive certificate trees, hostile verification corpora, tree-to-RPLAN package binding, and pseudo-Boolean proof requests. It also implements hash-chained ceremony genesis, announce, verify, and halt transitions; prefix verification does not require unreleased cuts. A connected Rhode Island RCTX contains all 25,649 Census blocks and applies the established deterministic island rule. The project has not yet generated and independently checked the three external proofs required for a full State cut, and no national exact-readiness claim is made. Internal review is not external peer review or legislative acceptance.

CONCLUSION

The revised proposal replaces “one algorithm draws the map” with “one evidence standard governs the process.” This is less mechanically absolute and more institutionally defensible. Its value is not removal of politics but precommitment, reproducibility, visible alternatives, and reviewable reasons.

REFERENCES

Rucho v. common cause, 2019. 588 U.S. 684 (2019). Partisan gerrymandering presents political questions beyond federal court reach; state courts retain jurisdiction. Creates legislative opening for the DIA.